



GET • INSTALL • RUN

Model Splitter

A clear, illustrated guide to downloading, installing, launching, and updating the application.

Browser + native desktop workflows

Browser: macOS · Windows · Linux
Desktop: macOS 11.3 or newer

INSTALLATION GUIDE

Edition 1.0 · July 2026
Application version 0.1.0

START HERE

Choose your route

Model Splitter runs from the same source code in two ways. Browser mode is the shortest path; native mode adds a macOS window and native file dialogs.

FASTEST

Browser mode

Best for trying the app or using it on macOS, Windows, or Linux.

Needs: Node.js + npm

Launch: `npm run dev`

MACOS

Native desktop mode

Best for a desktop window, native open/save dialogs, and building a DMG.

Also needs: Xcode tools + Rust

Launch: `npm run tauri:dev`

Source distribution. There is no prebuilt installer in the repository yet. Both routes begin by downloading the source and installing its dependencies.

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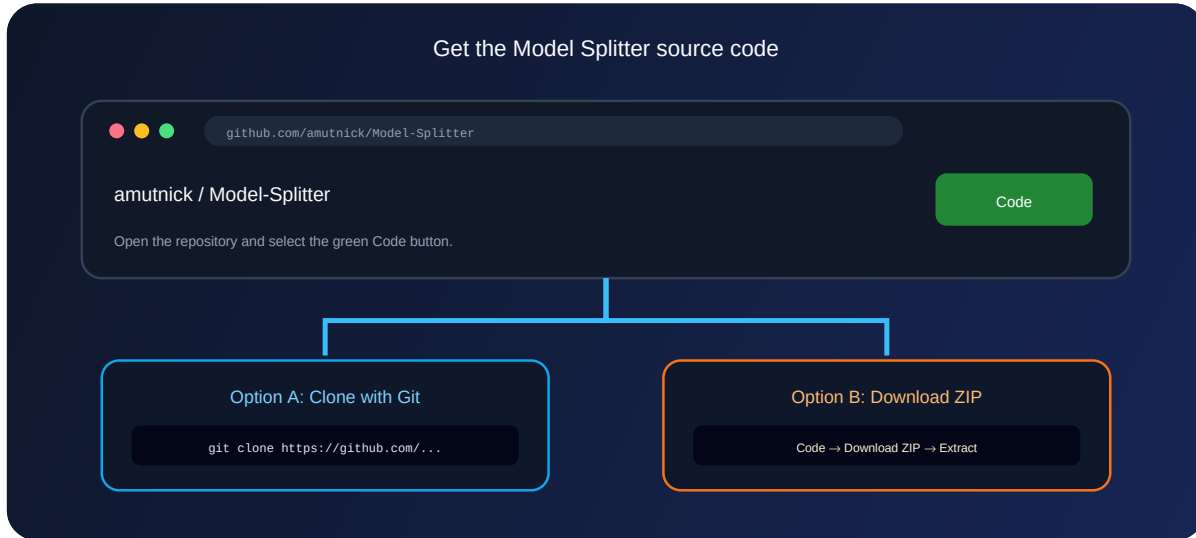
Requirements at a glance

Run mode	Operating system	Required software
Browser	macOS, Windows, or Linux	Node.js and npm
Native desktop	macOS 11.3+	Node.js, npm, Xcode tools, and Rust

STEP 1

Get the source code

Open github.com/amutnick/Model-Splitter, select the green **Code** button, and choose one download method.

**RECOMMENDED**

A. Clone with Git

Install [Git](#), then open Terminal or PowerShell.

```
git clone https://github.com/amutnick/Model-Splitter.git
cd Model-Splitter
```

This is the easiest route for future updates.

NO GIT NEEDED

B. Download ZIP

- 1 Select **Code** → **Download ZIP**.
- 2 Extract the downloaded file.
- 3 Open a terminal in the extracted folder.

```
# macOS / Linux
cd ~/Downloads/Model-Splitter-main

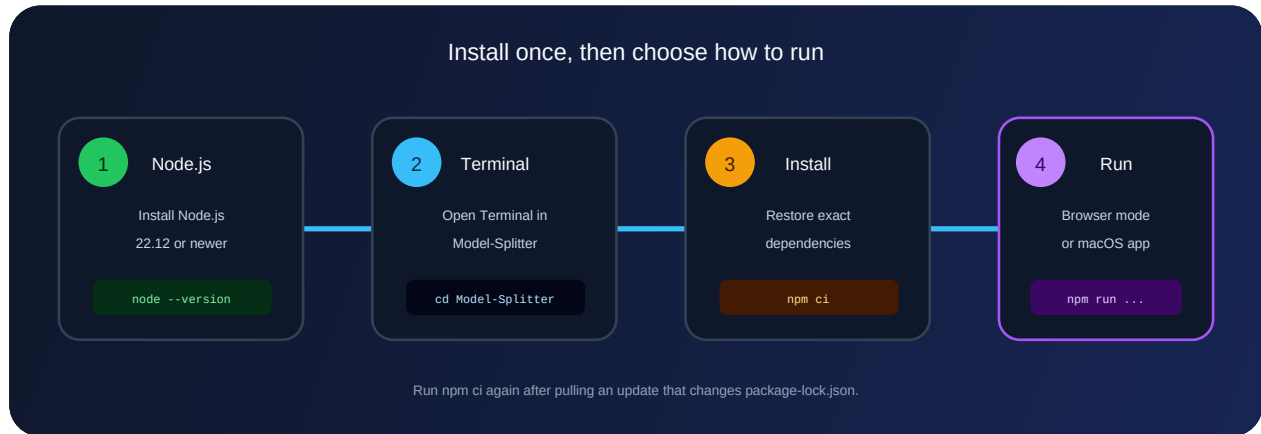
# Windows PowerShell
cd "$HOME\Downloads\Model-Splitter-main"
```

Easy folder shortcut. Type `cd` with a trailing space, drag the project folder into the terminal window, and press Return/Enter.

STEPS 2-3

Install the prerequisites

Install Node.js once, then restore Model Splitter's exact dependency versions inside the project folder.



2 • Install Node.js

- 1 Visit nodejs.org.
- 2 Download a current **LTS** installer.
- 3 Run it with the default options.
- 4 Close and reopen the terminal.

Supported: Node 20.19+ within 20.x, or Node 22.12+.

```
node --version  
npm --version
```

3 • Install dependencies

Confirm the terminal is in **Model-Splitter** or **Model-Splitter-main**, then run:

```
npm ci
```

npm reads `package-lock.json` and creates a local `node_modules` folder.

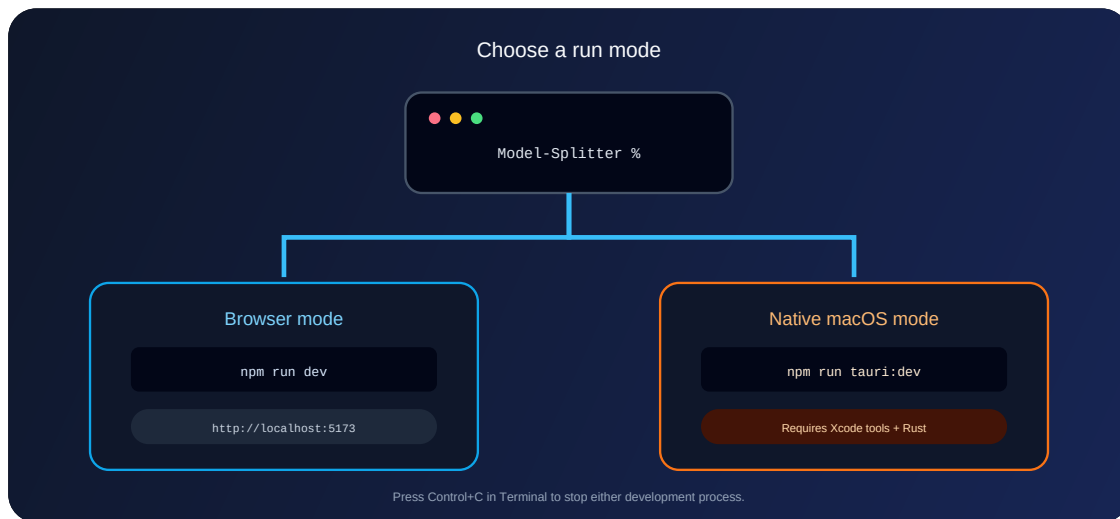
A successful install returns to the prompt without a red error.

Command not found? Restart the computer or reinstall Node.js with its “Add to PATH” option enabled, then check the version commands again.

STEP 4 • CHOICE A

Run in a browser

This is the fastest route and works on macOS, Windows, and Linux.



1

Start the development server

Run this in the project folder:

```
npm run dev
```

2

Wait for the local address

Vite prints an address when it is ready:

```
VITE ready
Local: http://localhost:5173/
```

3

Open the app

Visit <http://localhost:5173> in Chrome, Edge, Firefox, or Safari.

4

Keep Terminal open

The terminal process is serving the application.

5

Stop when finished

Return to Terminal and press **Control+C**.

Production check

Validate TypeScript and create an optimized build.

```
npm run build
```

Preview the build

Serve the generated production files locally.

```
npm run preview
```

STEP 4 • CHOICE B

Run the native macOS app

Native mode requires macOS 11.3 or newer. Complete these one-time platform setup steps before launching it.

1 • Install Xcode command-line tools

```
xcode-select --install
```

Follow the macOS installer prompts. If the tools are already installed, continue.

2 • Install Rust

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
```

Choose the default installation, reopen Terminal, and verify:

```
source "$HOME/.cargo/env"  
rustc --version  
cargo --version
```

3 • Launch Model Splitter

```
npm run tauri:dev
```

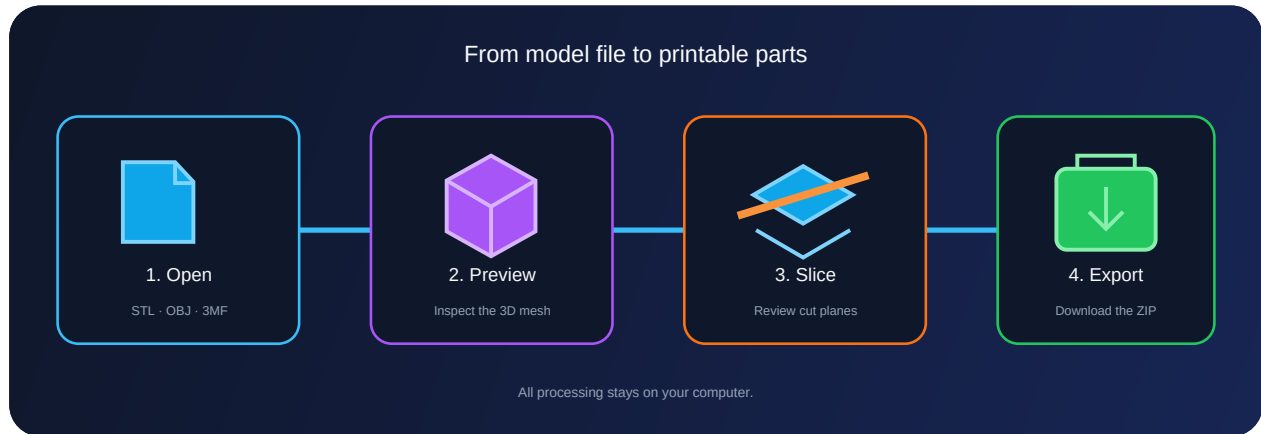
The first launch downloads and compiles Rust dependencies, so it can take several minutes. Later launches are much faster.

Expected result. A native window titled Model Splitter opens. Press Control+C in Terminal to stop the development process.

STEP 5

Use Model Splitter

The browser and native versions share the same modeling workflow. All geometry processing stays on your computer.



- 1 Select **Browse Files** or drop a model into the window.
- 2 Inspect the model and its solid/open status.
- 3 Enable **Repair open mesh** if simple boundary holes need filling.
- 4 Choose a segmentation strategy and adjust its options.
- 1 Select **Slice Model**.
- 2 Enable **PEG** on selected cuts that need guide connectors.
- 3 Review the generated parts, sockets, and cut planes.
- 4 Select **Download .ZIP** to export segmented STL files.

Supported input

STL

Binary or ASCII

OBJ + MTL

Wavefront mesh

3MF

Assemblies + colours

Saving differs by mode. Browser mode downloads through the browser. Native mode opens the macOS Save dialog.

STEP 6 • OPTIONAL

Build and install the macOS app

Create a reusable application and disk image so Model Splitter can launch without a development command.

1 Build on a Mac

Create the release bundle:

```
npm run tauri:build
```

2 Open the generated bundle folder

Reveal the results in Finder:

```
open src-tauri/target/release/bundle
```

3 Open the DMG

Enter the `dmg` folder and double-click the generated disk image.

4 Install

Drag Model Splitter into the Applications folder.

5 Launch

Open Model Splitter from Applications.

Where the files go

```
src-tauri/  
└─ target/  
   └─ release/  
      └─ bundle/  
         ├── macos/ ← Model Splitter.app  
         └─ dmg/   ← installable disk image
```

About macOS security. Local builds are not automatically signed or notarized. If macOS blocks your own build, Control-click the app in Finder, choose **Open**, and confirm. Never bypass a warning for an app from an untrusted source.

Platform note. A macOS `.app` or `.dmg` must be built on macOS. Windows and Linux users should use browser mode.

KEEP IT CURRENT

Update and troubleshoot

Update the project using the same method you used to download it, then use the quick fixes below for common setup problems.

Git clone

Update the existing folder:

```
git pull
npm ci
```

Downloaded ZIP

Download and extract a fresh ZIP, open its folder, then run:

```
npm ci
```

Common problems

Problem	What to do
node or npm not found	Reinstall Node.js LTS, reopen Terminal, and check <code>node --version</code> .
npm ci network error	Check the internet connection, VPN, proxy, or firewall, then retry.
Port 5173 is in use	Stop the old Vite process with Control+C, close old terminal sessions, and retry.
Browser does not open	Open http://localhost:5173 manually.
cargo not found	Reopen Terminal or run <code>source "\$HOME/.cargo/env"</code> .
Xcode or linker error	Run <code>xcode-select --install</code> , finish the installer, and retry.
Native build on Windows/Linux	Use browser mode. Build the app and DMG on macOS.
3D view is blank or slow	Update the browser and graphics drivers, enable hardware acceleration, or try a smaller model.
macOS blocks the app	Control-click and choose Open only if you built the app yourself.

Still stuck? Copy the complete terminal error—not only its final line—when asking for help. It usually identifies the missing tool or failed step.

REFERENCE

Commands at a glance

Run these commands from the project folder.

```
npm ci
```

Install the exact JavaScript dependencies.

```
npm run dev
```

Start browser development mode at localhost:5173.

```
npm run build
```

Type-check and create the optimized web build.

```
npm run preview
```

Preview the generated web build.

```
npm run tauri:dev
```

Compile and launch the native macOS development app.

```
npm run tauri:build
```

Create the native app and DMG release bundle.

Clean reinstall · macOS/Linux

```
rm -rf node_modules  
npm ci
```

Clean reinstall · PowerShell

```
Remove-Item -Recurse -Force node_modules  
npm ci
```

Ready-to-run checklist

- The project folder is downloaded or cloned.
- `node --version` and `npm --version` work.
- `npm ci` completes without an error.
- Browser mode opens localhost:5173—or native prerequisites are installed.
- An STL, OBJ, or 3MF model is ready to load.

More information. Repository: github.com/amutnick/Model-Splitter
Developer-focused native notes: [TAURI_SETUP.md](#)

Model Splitter · local-first STL, OBJ, and 3MF feature slicing for multi-colour 3D printing